Supp Q 1

$$Vol = \frac{\pi r^2 h}{3} \qquad SA = \pi r \sqrt{h^2+r^2}$$

$$\nabla vol = \left(\frac{\pi r^2}{3}, \frac{2h\pi r}{3}\right) \qquad Vol = 1$$

$$\nabla SA = \left(\frac{h\pi r}{\sqrt{h^2+r^2}}, \frac{\pi r^2}{\sqrt{h^2+r^2}} + \pi\sqrt{h^2+r^2}\right)$$

Need to solve $\nabla vol = \lambda \cdot \nabla SA \qquad Vol = 1$

$$\frac{\pi r^2}{3} = \frac{\lambda \cdot h\pi r}{\sqrt{h^2+r^2}} \qquad \frac{\pi r^2 h}{3} = 1$$

$$\frac{2h\pi r}{3} = \left(\frac{\pi r^2}{\sqrt{h^2+r^2}} + \pi\sqrt{h^2+r^2}\right)\lambda$$

$$h = \frac{3}{\pi r^2} \qquad \lambda = \frac{r}{3h}\sqrt{h^2+r^2}$$

$$h^2 = \frac{9}{\pi^2 r^4}$$

Subbing into Remaining eqn

SQ1

2

$$\frac{2h\cancel{\pi}\cancel{r}}{\cancel{3}} = \left(\frac{\cancel{\pi}\, r^2}{\cancel{\sqrt{h^2+r^2}}} + \cancel{\pi}\sqrt{r^2+h^2}\right)\frac{\cancel{r}}{\cancel{3}h}\sqrt{h^2+r^2}$$

$$\Rightarrow 2h^2 = r^2 + r^2 + h^2$$

$$h^2 = 2r^2$$

$$\Rightarrow \frac{9}{\pi^2 r^4} = 2r^2 \quad \Rightarrow \quad r = \left(\frac{9}{2\pi^2}\right)^{1/6} \text{cm}$$

$$h = \frac{3}{\pi}\left(\frac{2\pi^2}{9}\right)^{1/3} \text{cm}$$

Supp Q 2

a)
$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{2\pi}\int_{-\pi/2}^{\pi/2} \tfrac{1}{2}\,dx$$

$$= \frac{1}{2\pi}\left(\frac{\pi}{4}+\frac{\pi}{4}\right) = \frac{1}{4}$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx = \frac{1}{\pi}\int_{-\pi/2}^{\pi/2} \tfrac{1}{2}\cos(nx)\,dx$$

$$= \frac{1}{2\pi n}\left[\sin\left(\frac{n\pi}{2}\right) + \sin\left(\frac{n\pi}{2}\right)\right] = \frac{\sin\left(\frac{n\pi}{2}\right)}{\pi n}$$

$$= \frac{(-1)^k}{\pi(2k+1)} \qquad k = 0, 1, 2, \ldots$$

So Fourier series is

$$\frac{1}{4} + \frac{1}{\pi}\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)}\cos(nx)$$

b) $$\frac{1}{4} + \frac{1}{\pi}\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k+1)} = \frac{1}{2} \Rightarrow$$

Set $x = 0$ $$\boxed{\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k+1)} = \frac{\pi}{4}}$$

Supp Q3

The solution is

$$e^{-4t}\cos(2x) + e^{-9t}\cos(3x) = u(x,t)$$

Supp Q4

$\nabla h = (-\sin x, \cos y) = (0,0) \Rightarrow$

$x = k\pi \quad y = (\frac{1}{2} + l)\pi \quad k, l \in \mathbb{Z}$

So cps are points $\pi(k, \frac{1}{2}+l)$, $k, l \in \mathbb{Z}$

$$D^2 h = \begin{pmatrix} -\cos x & 0 \\ 0 & -\sin y \end{pmatrix}\Bigg|_{\pi(k, \frac{1}{2}+l)} \quad k, l \in \mathbb{Z}$$

$$= \begin{pmatrix} (-1)^{k+1} & 0 \\ 0 & (-1)^{l+1} \end{pmatrix}$$ So

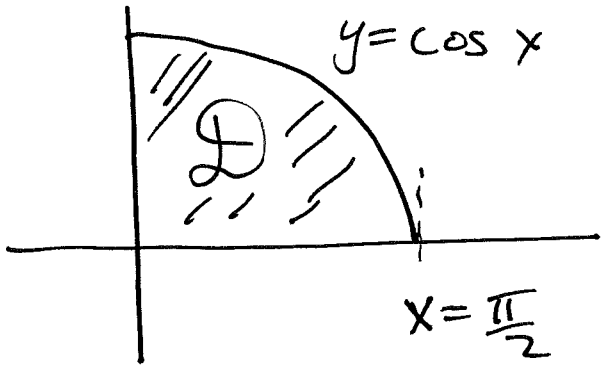
if k is even, l is odd $\rightarrow$ cp is a saddle

k is odd l is even $\Rightarrow$ cp is a saddle

k is even l is even $\Rightarrow$ cp is a ~~saddle~~ local max

k is odd l is odd $\Rightarrow$ cp is a ~~saddle~~ local min

Supp Q5



$\rho = x + y$

Centre of mass is $(\bar{x}, \bar{y})$ where

$$\bar{x} = \frac{\iint_D x\rho(x,y)\,dx\,dy}{\iint_D \rho(x,y)\,dx\,dy} \qquad \bar{y} = \frac{\iint_D y\,\rho(x,y)\,dx\,dy}{\iint_D \rho(x,y)\,dx\,dy}$$

$$\iint_D \rho(x,y) = \int_0^{\frac{\pi}{2}} \int_0^{\cos x} (x+y)\,dx\,dy =$$

$$\int_0^{\frac{\pi}{2}} xy + y^2/2 \Big|_0^{\cos x} dx = \int_0^{\pi/2} x\cos x + \tfrac{1}{2}\cos^2 x\,dx$$

$$= \frac{x}{4} + \cos x + x\sin x + \frac{1}{8}\sin(2x) \Big|_0^{\pi/2}$$

$$= \frac{\pi}{8} - 1 + \frac{\pi}{2} = \boxed{\frac{5\pi}{8} - 1}$$

Supp Q5

$$\iint_D x(x+y)\,dy\,dx = \int_0^{\frac{\pi}{2}} \int_0^{\cos x} x^2 + xy \; dy\,dx$$

$$\int_0^{\frac{\pi}{2}} \int_0^{\cos x} x^2 + xy \; dy\,dx = \int_0^{\pi/2} x^2 y + xy^2 \Big|_0^{\cos x} dx$$

$$= \int_0^{\frac{\pi}{2}} x^2 \cos x + x\cos^2 x \; dx =$$

$$\frac{x^2}{4} + 2x\cos x + \frac{1}{8}\cos(2x) + (x^2-2)\sin x + \frac{x}{4}\sin(2x)\Big|_0^{\frac{\pi}{2}}$$

$$= \frac{\pi^2}{16} + 0 - \frac{1}{8} + \left(\frac{\pi^2}{4} - 2\right) + 0 - 0 - 0 - \frac{1}{8}$$

$$- 0 - 0 = \boxed{\frac{5\pi^2}{16} - \frac{9}{4}}$$

Supp Q5

$$\int_0^{\pi/2}\int_0^{\cos x} xy + y^2 \, dy\, dx =$$

$$\int_0^{\pi/2} \frac{xy^2}{2} + y^3/3 \Big|_0^{\cos x} dx =$$

$$\int_0^{\pi/2} \frac{x\cos^2 x}{2} + \frac{\cos^3 x}{3} =$$

$$\frac{x^2}{4} + \frac{1}{8}\cos(2x) + \frac{1}{4}\sin x + \frac{x}{4}\sin(2x) + \frac{1}{36}\sin(3x)\Big|_0^{\frac{\pi}{2}}$$

$$= \frac{\pi^2}{16} - \frac{1}{8} + \frac{1}{4} + 0 - \frac{1}{36} - 0 - \frac{1}{8} - 0 - 0 - 0$$

$$= \frac{\pi^2}{16} - \frac{1}{36}$$

So the centre of mass is $(\bar{x}, \bar{y})$

$$\bar{x} = \left(\frac{5\pi^2}{16} - \frac{9}{4}\right) \Big/ \left(\frac{5\pi}{8} - 1\right)$$

$$\bar{y} = \left(\pi^2/16 - \frac{1}{36}\right) \Big/ \left(5\pi/8 - 1\right)$$

Supp Q6

$y_p = Ae^{-x}$

$y_p'' = Ae^{-x}$

$y_p'' + y_p = 2Ae^{-x} = e^{-x} \Rightarrow A = 1$

Solution to homogeneous part is

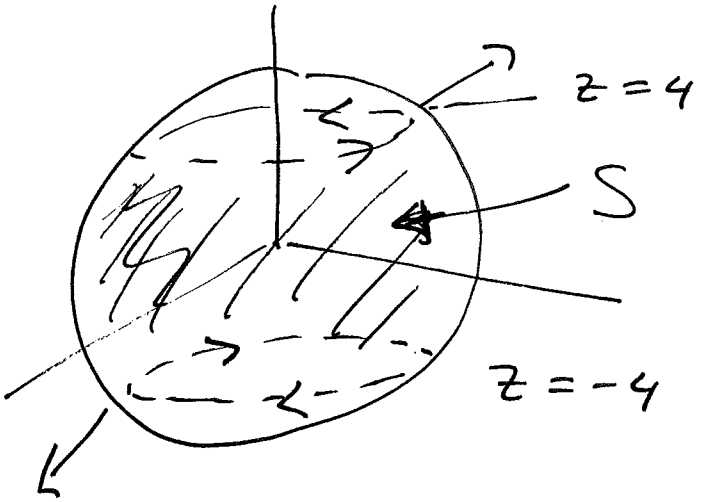
$y = c_1 \cos x + c_2 \sin x$

General soln is

$\boxed{c_1 \cos x + c_2 \sin x + e^{-x}}$

Supp Q7

a)



∂S is 2 circles $\quad \gamma_1 = (\cos t, \sin t, 4)$

$\gamma_2 = (\cos t, -\sin t, 4)$

b)

$$f = (yz + 2x, \; -xz + 4y^2, \; z^4 - 2)$$

$$\nabla \times f = \begin{vmatrix} i & j & k \\ \partial x & \partial y & \partial z \\ yz + 2x & -xz + 4y & z^4 - 2 \end{vmatrix}$$

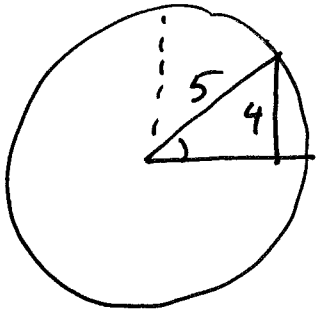
$$\nabla \times f = (x, y, -(x+y)) \qquad \vec{n} = (x, y, z)$$

$$\iint_S (\nabla \times f) \cdot \vec{n} \, dS = \iint_S (x^2 + y^2 - z(x+y) \, dS$$

Supp Q7

Can Parametrize S in Spherical coordinates

$$x = 5\cos\theta\sin\varphi$$
$$y = 5\sin\theta\sin\varphi$$
$$z = 5\cos\varphi$$



$\frac{\pi}{2} = \varphi$

$$S = (\theta, \varphi) \in [0, 2\pi] \times \Big[\underbrace{-\mathrm{Arctan}\left(\tfrac{4}{5}\right) + \tfrac{\pi}{2}}_{\varphi_1}, \underbrace{\mathrm{Arctan}\left(\tfrac{4}{5}\right) + \tfrac{\pi}{2}}_{\varphi_2}\Big]$$

$$\iint_S (\nabla \times f)\cdot\vec{n}\, dS = \int_{\varphi_1}^{\varphi_2}\int_0^{2\pi} 25\left(\sin^2\varphi - \frac{\sin(2\varphi)}{2}(\cos\theta + \sin\theta)\right) dS$$

$$dS = \begin{vmatrix} -5\sin\theta\sin\varphi & 5\cos\theta\cos\varphi \\ 5\cos\theta\sin\varphi & 5\sin\theta\cos\varphi \end{vmatrix}$$

$$= \frac{25}{2}\sin 2\varphi\, d\theta\, d\varphi$$

So Integral is

$$\int_{\varphi_1}^{\varphi_2}\int_0^{2\pi} \frac{625}{2}\left(\sin^2\varphi\sin 2\varphi - \sin^2(2\varphi)(\cos\theta + \sin\theta)\right) d\theta\, d\varphi$$

Sup Q7

$$= \int_{\varphi_1}^{\varphi_2} \frac{625}{2} \sin^2\varphi \sin 2\varphi \, d\varphi$$

$$= \int_{\frac{\pi}{2} - \mathrm{Atan}(\frac{4}{5})}^{\frac{\pi}{2} + \mathrm{Atan}(4/5)} 625 \cos(\varphi) \sin^3(\varphi) \, d\varphi$$

~~[illegible]~~ write $\xi = \varphi - \frac{\pi}{2}$

$$= \int_{-\mathrm{Atan}(4/5)}^{\mathrm{Atan}(4/5)} 625 \sin(\xi) \cos^3(\xi) \, d\xi$$

This fn is odd $\Rightarrow$ $\boxed{\int = 0.}$

<u>Rk</u> this can also be done via Stoke's Thm - this is how I did it originally

Supp Q8

d'Alembert's soln is

$$\frac{1}{2}\left(f(x+ct) + f(x-ct)\right) + \frac{1}{2c}\int_{x-ct}^{x+ct} g(s)\,ds$$

$c = 2$

$$f(x) = \frac{1}{(3-x)^2+4x} = \frac{1}{x^2-6x+9+4x} = \frac{1}{x^2-2x+9}$$

$$\int g(s) = \text{Arctan}(s)$$

So the solution is

$$\boxed{\frac{1}{2}\left[\frac{1}{(x+2t)^2-2(x+2t)+9} + \frac{1}{(x-2t)^2-2(x-2t)+9}\right] + \frac{1}{2}\left(\text{Arctan}(x+2t) - \text{Arctan}(x-2t)\right)}$$

$= u(x,t)$

Supp Q9

$$SA = \iint_{D_a} \sqrt{1 + f_x^2 + f_y^2}\, dx\,dy$$

$$f = x^2 + y^2 \qquad f_x = 2x \qquad f_y = 2y$$

$$= \iint_{D_a} \sqrt{1 + 4x^2 + 4y^2}\, dx\,dy$$

Polar $x = r\cos\theta$
$y = r\sin\theta$

$dx\,dy = r\, dr\, d\theta$

$$\int_0^{2\pi} \int_0^a r\left(\sqrt{1+4r^2}\right) dr\, d\theta =$$

$$2\pi \int_0^a \left(r\sqrt{1+4r^2}\right) dr = \frac{2\pi}{8} \int_1^{1+4a^2} u^{1/2}\, du$$

$u = 1 + 4r^2 \qquad a \to 1 + 4a^2$

$du = 8r\, dr \qquad 0 \to 1$

$dr = \dfrac{du}{8r}$

$$= \frac{\pi}{4} \cdot \frac{2}{3} u^{3/2} \Big|_1^{1+4a^2} = \frac{\pi}{6}\left((1+4a^2)^{3/2} - 1 \right)$$

Want $= 1$

Supp Q9

$$\Rightarrow \frac{\pi}{6}\left((1+4a^2)^{3/2}-1\right)=1$$

$$\Rightarrow \boxed{a = \frac{1}{2}\left[\left(\frac{6}{\pi}+1\right)^{2/3}-1\right]^{1/2}}$$